Replication of Study 2 by Markowitz (2024, PNAS Nexus)

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Table of contents

Introduction . . . 1
Methods . . . 3
- Power Analysis . . . 3
- Planned Sample . . . 4
- Materials . . . 4
- Procedure . . . 4
- Analysis Plan . . . 4
- Differences from Original Study . . . 4
- Methods Addendum (Post Data Collection) . . . 4

Results . . . 5
- Data preparation . . . 5
- Confirmatory analysis . . . 5
- Exploratory analyses . . . 5

Discussion . . . 5
- Summary of Replication Attempt . . . 5
- Commentary . . . 5

Introduction

One research area I am interested in is understanding the psychological implications of generative AI, including how people perceive and respond to AI-generated information. I am particularly interested in whether using AI to make information more accessible changes not only people's understanding of that information but also their perceptions of its source, the author. Scientific information is one area where information may be difficult to grasp for non-experts because the language can be complex. I therefore chose to conceptually replicate

Markowitz's paper (2024), titled "From Complexity to Clarity: How AI Enhances Perceptions of Scientists and the Public's Understanding of Science." In Study 2, Markowitz experimentally tested whether reading simpler AI-generated scientific summaries, rather than more complex human-written summaries, affected perceptions of scientists' credibility, trustworthiness, and intelligence. In Study 3, Markowitz expanded this study to have participants read more summaries and complete one multiple choice question to capture their understanding of the overall gist of the summary. This experiment aligns with my broader interest in how generative AI changes people's psychological responses to information and those who communicate it.

Study 1 essentially consisted of gathering the PNAS significance statements, generating AI statements, and then systematically comparing the linguistic differences, including common words, analytic writing, and readability. Then, in Study 2, the researcher selected five pairs of human-written PNAS significance statements and GPT-generated significance statements that showed the largest differences in common words from Study 1. The data file in the supplementary materials shows the two conditions coded as 'simple' and 'complex'. Of the five stimulus pairs, each pair having a human-written summary and an AI-generated summary, participants were randomly assigned to read three individual summaries that were randomly pulled from three out of the five pairs. Stated another way, each participant had the possibility of seeing three entirely AI-generated summaries or entirely human-written summaries from the way the original study operated. After reading each text, participants would complete manipulation checks assessing its clarity, complexity, and how much of the summary they understood. Participants would then assess the "scientist's" intelligence, credibility, and trustworthiness and indicate whether they believed the summary was written by a human or by AI. It was unclear in the original study whether AI disclosure of the generated summaries occurred. The order of the measures and the items within each block would be randomized, as the original paper had done. Basic demographic information would also be collected.

To address some comments from the PSYC 201A teaching team, I believe that it would be useful to randomly assign the AI-generated or human written statements dependently, in balancing the versions within participants so that no participant gets either all human-written statements or AI-generated statements. I also think it would be fruitful to include an AI-disclosure condition to assess how this label changes the way that participants perceive the statements, such as completely dismissing the statement because it is AI-generated. I plan to add the AI disclosure condtion to make the study a between subjects design so that I can still assess the orignal study as a direct replication. As for concerns about the time frame for completing this study, I agree and I think that adding more summaries that participants will assess would add to the amount of time participants would spend on this study. Therefore, I plan to also include Study 3's perception and comprehension measures. To ensure the time participants spend on this study is enough to grant them SONA credits (30 minutes for 0.25 credits), I will need to run through the study myself and see. If the study is still short on time, I plan to add more summaries that participants will read and assess. Another point from the PSYC 201A teaching team was to focus on one key test. I plan to run a simulation-based power analysis to determine which test to run given the original study's effect sizes.

To analyze this data and reproduce the original findings, I will employ linear mixed-effects models because each participant provides multiple observations and individual stimuli are presented to multiple participants. Futhor following Markowitz (2024), the models would include random intercepts for both participant and stimulus, with text condition (GPT/simple versus PNAS/complex) as the primary fixed effect.

One anticipated challenge will be to gather enough participant data in the time frame of a few weeks, depending on the sample size from the power analysis. Another challenge brought up by the teaching team was the sample population difference. The original study drew data from Prolific, who may not be familiar with AI-generated material. For this replication project, SONA participants consist of undergraduate students, who may be very familiar with AI tools and AI-generated outputs. Because of this familiarization with AI, SONA participants' assessments may lead to variations in their ratings and human/AI judgments. Another challenge is that generative AI tools have increasingly improved in their capabilities from 2024 to 2026. This means that the 2024 outputs might be more recognizable due to the nature of AI tool outputs at the time. Now, AI generated outputs may differ, where people may find it more difficult to distinguish AI-generated text from human-written text, because current models produce writing that is more varied and natural-sounding than the 2024 outputs. To stick to the original study, I plan to use Markowitz's original GPT-generated summaries as stimuli rather than generating new ones.

For my learning goals, I want to learn how to concretely run a power analysis. I am also interested in simulating the results of this study, especially since I need to do so anyways to determine the effect sizes since the original paper did not include them. I hope to develop a deeper understanding of coding in R. Regarding feasibility, I think if I stick with the orignal AI-generated summary statements, it will be feasible to replicate this experiment, except with more summary statements and adding an AI-disclosure condition.

Link to repo: https://github.com/sisegall/markowitz2024

Linked pathway to the original paper: https://github.com/sisegall/markowitz2024/blob/main/original_paper/Markowitz_2024_PNASNexus_AI_science_summaries.pdf

Original data and materials (OSF): https://osf.io/64am3/ · Study 2 data: https://osf.io/download/xjnfg/ · Stimuli: https://osf.io/download/fnh74/ · Preregistration (Study 2): https://aspredicted.org/C3K_T31

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.